

Adam Zhu

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EDUCATION

The University of Texas at Austin

Aug 2024 - Present

Bachelor of Science, Computer Science. GPA: 3.9/4.0. Expected Graduation - May 2028

WORK EXPERIENCE

USA Computing Olympiad Mentor (using C++, Java, Python)

Feb 2025 – Present

- Guided **Silver** students through advanced algorithmic techniques including **binary search** and **graphs**.
- Improved implementation efficiency, increasing students' average coding speed for Silver problems by **20%**.

Web Developer (using JavaScript, React, Next.js, HTML, CSS, Tailwind, PostgreSQL, Express.js)

Mar 2023 – Jan 2025

- Engineered **Arcane Odyssey Guides**, surpassing **3 million views** by optimizing frontend performance with **React** and **Tailwind CSS** and designing an algorithm that sorts **40 billion** combinations in under **2 seconds**.
- Designed and developed **Monkey Sense**, improving students' mental math speed by **200%** by building a question generator with a multiplayer duel system using **Express.js** backend and a **PostgreSQL** leaderboard database.

Roblox and Unity Game Developer (using C#, Lua, Unity, Roblox Studio, GitHub)

Mar 2020 – Oct 2024

- Published three Roblox titles (**Dig To India**, **Monster Survival**, **Cave Mining**) amassing **1M** visits and **10% 7-day** retention through data-driven design and robust server security by encapsulating client-server communication.
- Developed **Geometry Battleground**, **CogQuest**, **The Past and The Curious**, and **Together In Spirit** using Unity, implementing object-oriented methodology with a **4-tier** class hierarchy, reusing over **10,000** lines of code.
- Achieved **#1 Overall** and **#1 Gameplay** in UT EGaDS! High Score Game Jam, **#1 People's Choice** and **#2 Best Narrative** in Texas Game Jam, and 3rd place in **BPA Software Engineering** after presenting to industry engineers.

PROJECTS

Word Bomb AI Bot (using Python, NumPy, OpenCV, Tesseract OCR, Computer Vision)

Jan 2025 – Feb 2025

- Optimized real-time puzzle solving to under **2 seconds** by integrating **Computer Vision** with **Tesseract OCR**.
- Achieved **97%** text recognition accuracy by preprocessing image data using **NumPy** and **OpenCV**.

Tetris AI Bot (using Python, PyTorch, Reinforcement Learning, Vue.js, Flask)

Apr 2025 – Present

- Designed and trained an **AI** Tetris bot using **Reinforcement Learning**, tweaking reward to survive **20x** longer.
- Built a frontend using **Vue.js** that instantly displays actions, using a **Neural Network** in the **Flask** backend.x

Clash Royale AI Deck Rater (using Python, NumPy, Pandas, JavaScript, Keras, Machine Learning)

Oct 2024 – Nov 2024

- Improved model accuracy and reduced loss by **80%** through **neural network optimization** in **Keras**.
- Curated a dataset of **2,000+** entries through **web scraping** and **data preprocessing** with JavaScript.

LeetCode Discord Bot (using Python, GraphQL, PostgreSQL, Matplotlib, GitHub)

Sep 2024 – Oct 2024

- Worked with a team of **4**, building a bot that serves 100+ users using the **discord.py** library and **Matplotlib**.
- Reduced GraphQL API calls by **50%** using PostgreSQL caching while keeping refreshes under **3 seconds**.

AWARDS

International Collegiate Programming Contest Regionals (sponsored by Jane Street, using C++) – **5th place**

Online Competitions (using C++, Algorithms): **LeetCode Guardian** (top 0.5%), **Codeforces Candidate Master** (top 3%)

UT Dallas Battle of the Brains (sponsored by State Farm, using Java) – **1st Place** in Advanced Division

UIL State – CS (Java): 2x 1st Place Team, 3rd Individual; **Number Sense** (Mental Math): 4th Team, 8th Individual

SKILLS

- Languages:** Java, C++, JavaScript/TypeScript, Python, C#, SQL, Lua, HTML/CSS, GraphQL
- Frameworks:** React, Next.js, Node, Express, Tailwind, NumPy, Pandas, Keras, OpenCV, .NET
- Tools:** Git, GitHub, PostgreSQL, Unity, Unreal, WSL, Agile/SCRUM, REST APIs, Postman, Linux
- Topics:** Software Engineering, AI, Machine Learning, Computer Vision, Database, Full-Stack, Algorithm Design